Unit 4, Lesson 1: Plotting Integers Using One-Unit Scales for Axes

Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

Learning Target

☐ I can plot with integers using one-unit scales for axes.

4.1.1 Warm-Up: Guess the Number

An image of $\frac{1}{4}$ cup of candy corn is shown.

1. Estimate the number of pieces of candy corn that are in the measuring cup.

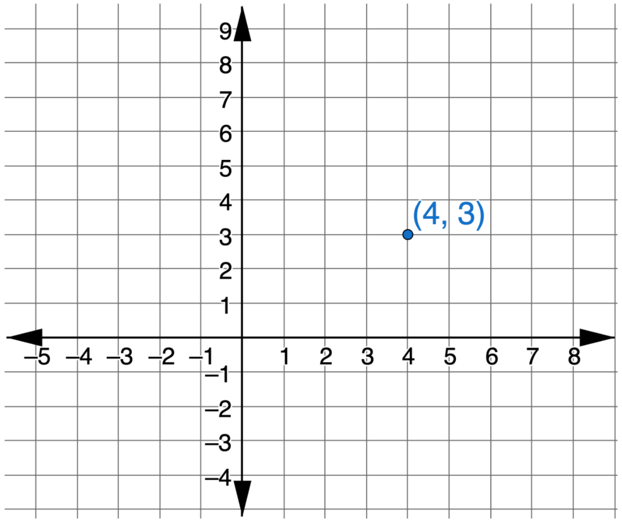
2. Ask two classmates the value they estimated. Record their value and write your summary of their reason. You are the only person who should write in the first two columns of the table. Have the person initial next to your summary stating that your summary is correct.

Guess	My Summary of Their Reason	Initials

3. How close was your estimate to the actual number?

4.1.2 Exploration: The Coordinate Plane

1. This graph shows a point that is plotted on a coordinate plane.



The two numbers are called the coordinates. The two numbers are also known as an ordered pair, (x, y).

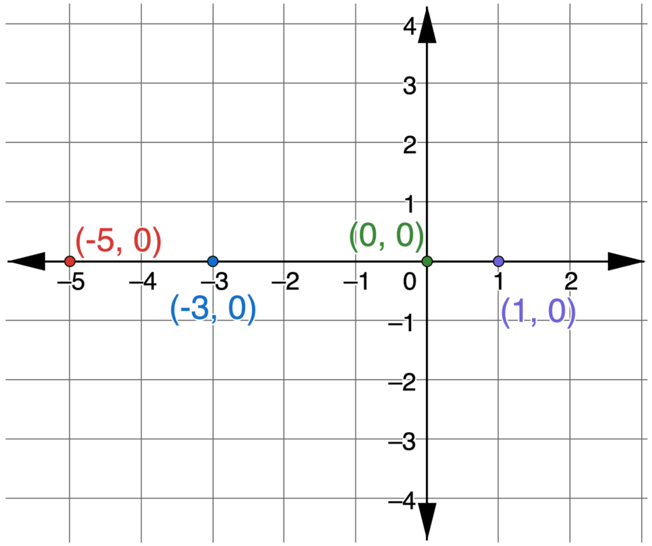
a. Discuss with your partner which coordinate will change if the point is dragged from left to right.

b. Discuss with your partner which coordinate will change if the point is dragged up or down.

c. If the point $(4, 3)$ could be dragged around on the coordinate grid, what changes would you expect to see in the ordered pair? Select all that apply.

- ☐ The x-coordinate becomes smaller when the point is moved to the left.
- ☐ The x-coordinate becomes larger when the point is moved to the left.
- ☐ The x-coordinate stays the same when the point is moved to the left.
- ☐ The y-coordinate becomes smaller when the point is moved up.
- ☐ The y-coordinate becomes smaller when the point is moved down.
- ☐ The y-coordinate stays the same when the point is moved up.

2. Four points are plotted on the coordinate grid.



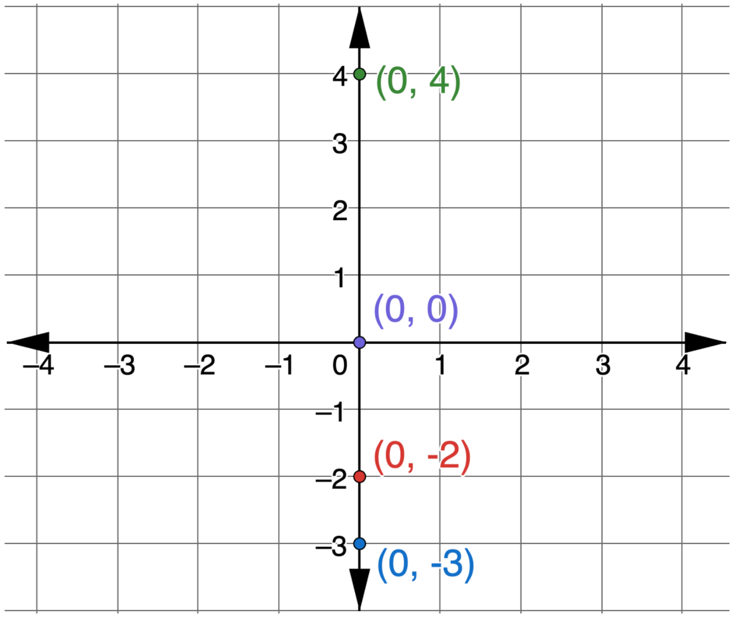
a. What do you notice about all four points?

b. What is the same about the points?

c. What is different about the points?

d. Discuss with your partner what is true about all the points on the x-axis.

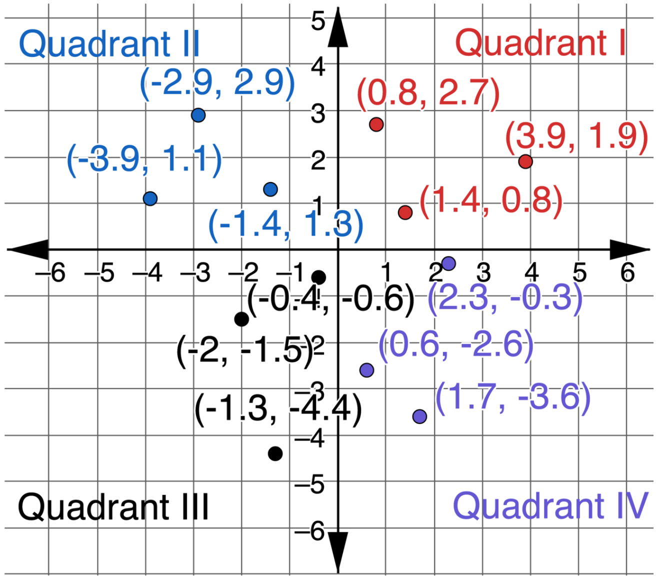
3. Four points are plotted on the coordinate grid.



a. What is the same about the points?

b. What is different about the points?

4. Twelve points are plotted on the coordinate grid. The grid shows the labels Quadrant I, Quadrant II, Quadrant III, and Quadrant IV.

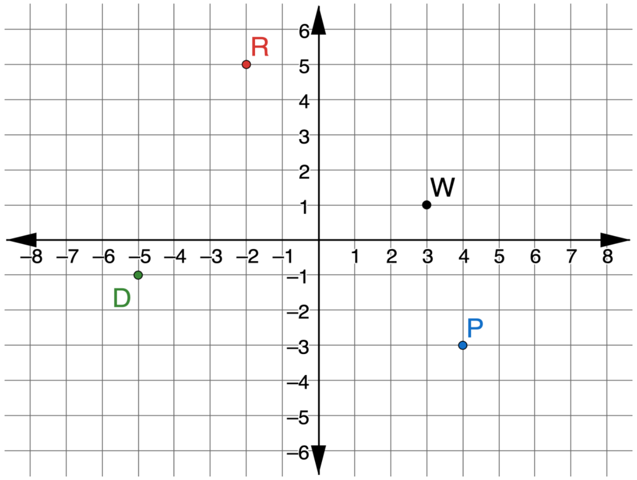


Work with your partner to summarize the characteristics of the points in each quadrant. Write your summary in the table.

Quadrant I	
Quadrant II	
Quadrant III	
Quadrant IV	

4.1.3 Bring It Together: Plotting Ordered Pairs

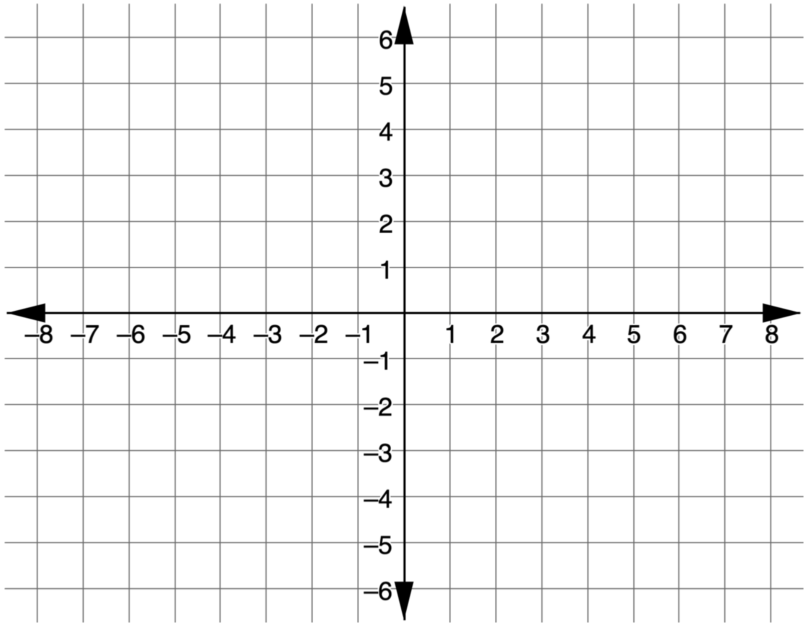
1. Points D, P, R, and W are plotted on the coordinate grid.



Select all of the descriptions of each point.

Point D	**Point P**
☐ The x-coordinate of D is -5.	☐ The x-coordinate of P is-3.
☐ The x-coordinate of D is -1.	☐ The x-coordinate of P is 4.
☐ The y-coordinate of D is -5.	☐ The y-coordinate of P is -3.
☐ The y-coordinate of D is -1.	☐ The y-coordinate of P is 4.
☐ Point D is in quadrant I.	☐ Point P is in quadrant I.
☐ Point D is in quadrant II.	☐ Point P is in quadrant II.
☐ Point D is in quadrant III.	☐ Point P is in quadrant III.
☐ Point D is in quadrant IV.	☐ Point P is in quadrant IV.
Point R	**Point W**
☐ The x-coordinate of R is -2.	☐ The x-coordinate of W is 1.
☐ The x-coordinate of R is 5.	☐ The x-coordinate of W is 3.
☐ The y-coordinate of R is -2	☐ The y-coordinate of W is 1.
☐ The y-coordinate of R is 5.	☐ The y-coordinate of W is 3.
☐ Point R is in quadrant I.	☐ Point W is in quadrant I.
☐ Point R is in quadrant II.	☐ Point W is in quadrant II.
☐ Point R is in quadrant III.	☐ Point W is in quadrant III.
☐ Point R is in quadrant IV.	☐ Point W is in quadrant IV.

2. Use the coordinate grid to plot and label points that have the characteristics described in the table.



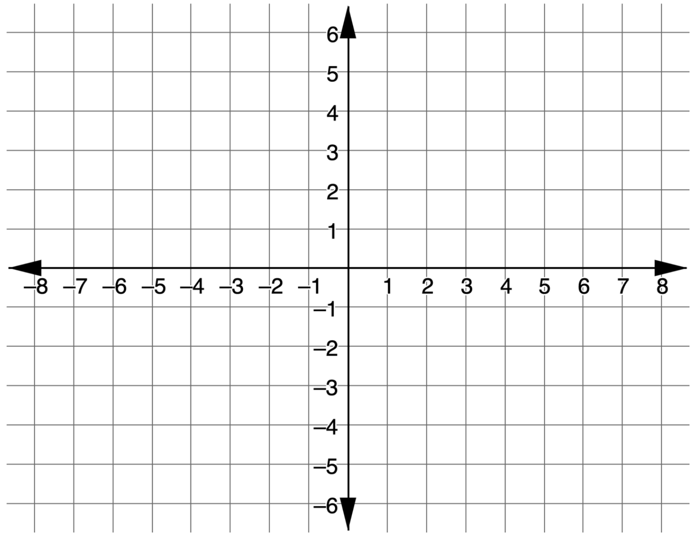
Point B has an x-coordinate of 0.	Point C has a y-coordinate of -4 and is in Quadrant III.
Point G has a y-coordinate of 0.	Point M has a y-coordinate of 2 and is in Quadrant I.
Point T has an x-coordinate of -3 and is in Quadrant II.	Point W has an x-coordinate of 1 and is in Quadrant IV.

4.1.4 Guided Instruction: Reflections

1. Use the points in the table for the following.

Point D $(1, 3)$	Point H $(1, -3)$	Point K $(-2, 4)$
Point R $(-5, -1)$	Point V $(5, -1)$	Point Z $(4, 4)$

a. Plot the ordered pairs from the table on the coordinate grid below. Label using the given letter.



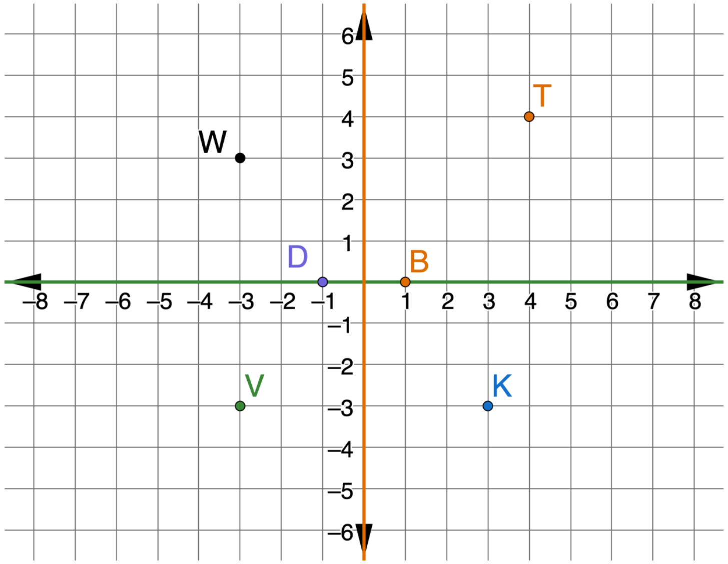
b. Select the pair of points that have the same x-coordinate.

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z

c. Select the pair of points that have a y-coordinate of -1.

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z

2. A coordinate grid with points D, B, K, T, V, and W is shown. The x-axis is green and the y-axis is orange.



a. Write the ordered pair of the reflection of Point V when the graph is folded across the x-axis.

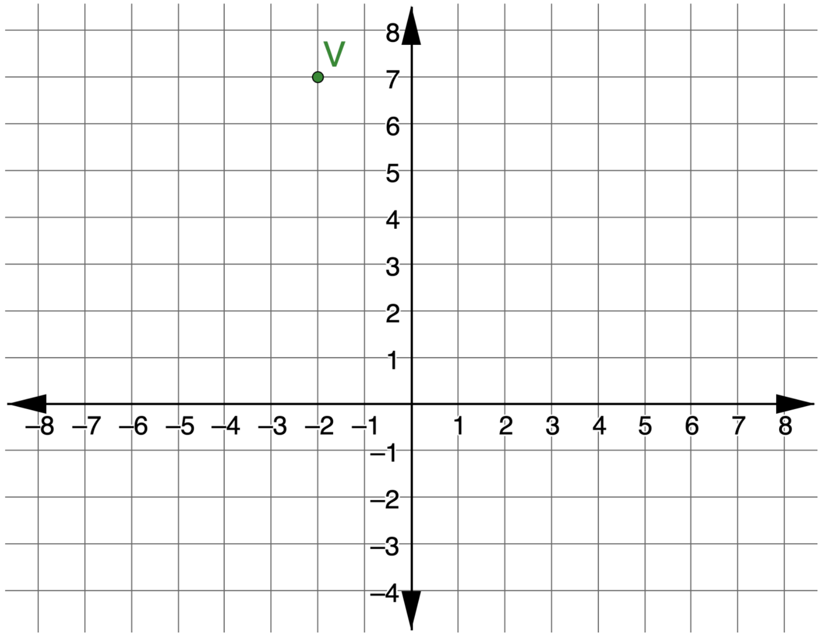
b. Write the ordered pair of the reflection of Point K when the graph is folded across the y-axis.

c. Write the ordered pair of the reflection of Point D when the graph is folded across the y-axis.

d. Write the ordered pair of the reflection of Point B when the graph is folded across the x-axis.

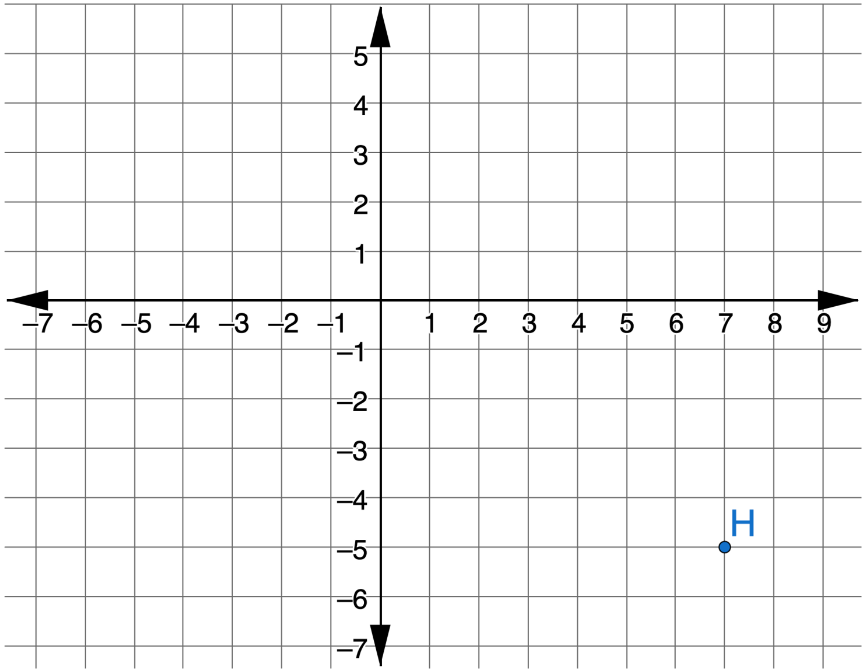
4.1.5 Guided Practice: Reflections

1. Point V is plotted on the coordinate grid.



What are the coordinates of the reflection of V across the y-axis?

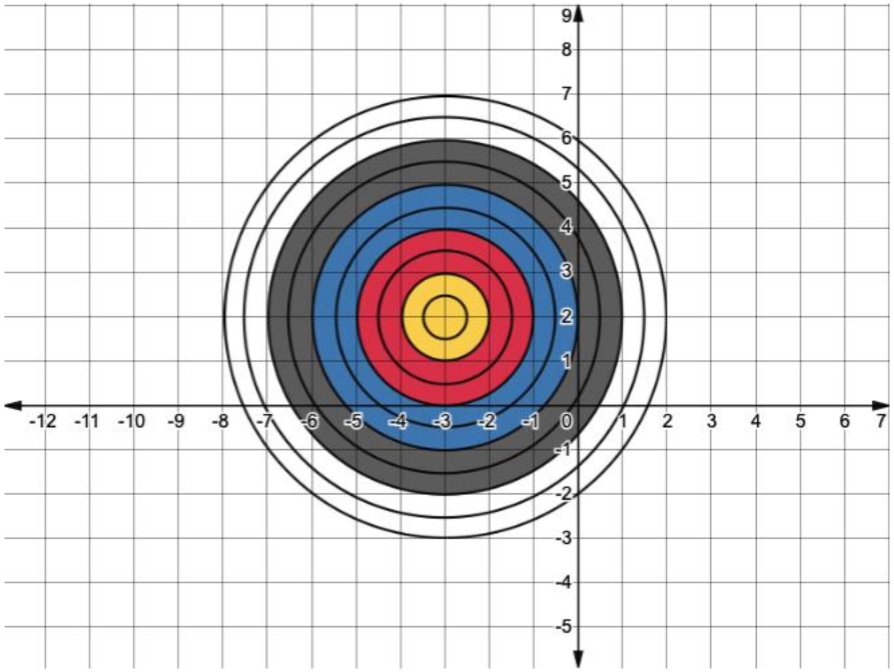
2. Point H is plotted on the coordinate grid.



What are the coordinates of the reflection of H across the x-axis?

4.1.6 Wrap-Up: Ticket Out the Door

A target is shown on the coordinate grid.



1. Write the ordered pair that will hit the bullseye.

2. A player throws two darts. The first dart hits the ordered pair $(-2, -1)$. The second dart hits the point that is the reflection of $(-2, -1)$ over the x-axis. Complete the table that describes the location of the second dart.

Color	Ordered Pair